MAGNUM CASH&CARRY IN AN AI-ENABLED MARKET

How should Magnum operate and compete? Use AI as a supervised drafting layer, and scale only on measured results.

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Company A: Carrefour · Company B: Magnum Cash&Carry · 25 October 2026

BOARD RECOMMENDATION

Use AI as a supervised drafting layer, and scale only on measured results

Recommendation. Magnum Cash&Carry should use AI as a supervised drafting and forecasting layer, not as a decision-maker: pilot demand forecasting and AI-drafted Пайда messages behind named human sign-off, adopt Carrefour-style data controls first, and scale only where measured shelf availability, forecast error and price accuracy beat today's process.

1 · PILOT FORECASTING

Reconcile stock records, then forecast one store and category. Scale to ten branches only at ≥98% availability and −10% forecast error.

2 · ADAPT AI COPY

AI drafts weekly Пайда messages from the public promo list; an automatic price check and the brand manager sign off every send.

3 · CONTROLS FIRST

Adopt data controls now; delay personalisation; reject autonomous ordering. People commit money and messages.

14.9 m ₸

12-month cost, 0.003% of 2024 revenue

≈16

branch-categories at target to cover running cost

First

Data boundary and stock reconciliation in months 0–3

Sources: business case (slide 12) from [2], [31], [36], [43]–[46]; verdicts from Weekly Records 3–9 and A1–A4.

Magnum is under real-revenue pressure and financially opaque, so tests must be cheap and stoppable

453.13 bn ₸
2024 revenue, Forbes Kazakhstan [31]

−13.1%
real change on 2023 (−5.6% nominal) [32], [35]

9,599
employees in 2024 [31]

230+
branches in 14 cities, company claim [36]

WEEK 1 ASSUMED

- Affordability matters: inflation 9.8% [4]; e-commerce 14.3% of retail [8]
- Ratios could be calculated in Week 8
- Uzbekistan entry (2021) as regional experience [14]
- AI opportunity = search and forecasting

WHAT I NOW KNOW

- Revenue fell in real terms in 2024; spending must be gated
- No public statements: the case must be proven on internal data
- Uzbekistan stores wound down [34]; Kaspi controls the e-commerce arm since 2023 [31]
- The weak touchpoint is promotion → purchase, not discount depth (Week 6)

Sources: [4], [8], [14], [31], [32], [34], [35], [36]. Real change deflates 2024 revenue by 8.6% inflation.

Every stakeholder obligation becomes a design rule in the operating model

Stakeholder	Obligation	Design rule in the model	Where
Customers	Accurate prices and available products	Price check + named sign-off; search shows checked stock only	§4, §6
Regulators	Law 94-V: consent, storage in Kazakhstan, transfer conditions [7]; AI Law [27]	Only public promo lists and aggregate sales enter AI tools; personalisation delayed	§5b, §6
Employees	Catch errors without pressure to approve	Training, recorded right to override, quarterly fallback drill	§5c
Suppliers	Accurate product facts, fair treatment	Supplier charter applied through Magnum rules [21], [22]	§3
Owners	Spending that stops when evidence fails	Stage gates with stop rules on every initiative	§8
Communities	Less food waste (SDG 12.3) [17]	Waste value is a pilot KPI	§5a, §7

External pressures: falling real revenue [31], [35] and routine clerical work as the most exposed to generative AI [40].

Sources: Weekly Record 2 stakeholder map; [7], [17], [21], [22], [27], [31], [35], [40].

The proposal protects how Magnum earns its margin; it adds no new revenue stream

UNCHANGED

- Value: availability and affordability for households and business buyers
- Capture: margin after wages, premises and logistics
- Governance: LLP general meeting decides each expansion [18]
- No paid AI service, no personalisation revenue

IMPROVED

- Fewer empty shelves recover lost sales (the only benefit counted in §7)
- Clearer promotions convert discount value already paid for (tracked, not monetised)
- Checked data cut waste and correction time

Conflict with §7 resolved: the business case counts recovered gross margin from availability only. Personalisation stays delayed, so no revenue from it is assumed.

Sources: Weekly Records 1 and 3, A1 §3; [18].

Magnum should take Carrefour's logic, not its scale

Practice at Carrefour	Verdict	Grounds for Magnum	Record
AI demand forecasting (SAS) [20]	PILOT	Proven elsewhere, not locally; stock records must reconcile first	W3–W5, A2
AI-written product copy, 2,000+ sheets [12]	ADAPT	Weekly Пайда messages with price check, Kazakh review, no personal data	W6, A3
Data controls after the CNIL fine [26]	ADOPT	Retention limits, AI notice and opt-out fit Law 94-V and the AI Law [7], [27]	W7, A3
Supplier ethics charter [21]	ADOPT	Applied through Magnum's corporate rules [22]	W3, A1
Hopla catalogue assistant [12], [29]	ADAPT	Kazakh and Russian questions, checked availability only, staff help kept	W1–W3, A1
AI agents in central functions [38]	ADAPT	Finance reporting on internal ledgers with the four checks	W8
Smart-store cameras and labels [37], [38]	DELAY	About €150 m capital programme; availability baseline and camera privacy first	Capstone
Personalisation and ChatGPT shopping [19]	DELAY	Consent, retention and labelling decisions first	W2, W3, W7, A3
Autonomous order approval	REJECT	A wrong recommendation must not become a purchase commitment	W3, A1

Sources: [7], [12], [19]–[22], [26], [27], [29], [37], [38]; verdicts from Weekly Records 1–8 and A1–A3.

AI drafts, people commit: one rule across five functions

Function	AI executes, human reviews	Human-only decision	Review → escalation	Fallback
Replenishment	Forecasts and draft quantities	Category manager approves orders	Planner, store staff → supply-chain lead; IT for faults	Manual order from physical count
Пайда promotions	Three draft messages from the public promo list	Brand manager approves the send	Copywriter, Kazakh reviewer → brand manager; legal for labelling	Approved message template
Finance reporting	Calculations and draft commentary	Finance manager signs the KPI report	Analyst applies four checks → finance manager	Manual monthly report
Catalogue search (months 9–12)	Answers with checked-availability items only	Service manager handles complaints	Store staff → digital product lead	Ordinary search and staff help
Training (HR)	Draft training material from pilot errors	HR lead approves curriculum	Category manager reviews content → HR lead	Classroom drill

Fixed-rule automation validates inputs and keeps logs in every row; Human + AI interpretation covers promotions and supplier constraints. Approval authority never moves to software.

AI-assisted visual Source: author’s consolidation of the Week 4 matrix and Weeks 5–9. Proposed roles; Magnum’s internal allocation is not public.

Reconcile records first, forecast second; the category manager still approves every order

Current bottleneck: stock and promotion records are reconciled by hand before quantities are set; during promotions recorded and shelf stock can differ (Weeks 3, 5).
Stop rule: if they still differ after reconciliation, forecasting moves to DELAY.

1 AUTOMATED
Validate sales and stock inputs; block stale data

→

2 AI + REVIEW
Forecast demand; draft quantities

→

3 HUMAN + AI
Adjust for promotions, delivery days, supplier limits

→

4 HUMAN ONLY
Category manager approves, changes or rejects

→

5 HUMAN + SYSTEM
Send approved order; receive goods

→

6 LOGGED
Check shelves and costs; review exceptions daily

≥98%
shelf availability (drives §7)

−10%
forecast error, WAPE

−10%
waste value

−10%
staff cost per order

AI-assisted visual Source: Week 5 process map (baseline to confirm with staff). Placeholder baseline: 8.3% world-average out-of-stock rate [43]. Carrefour tested SAS for 18 months [20].

Customers get clearer, checked offers; personalisation waits for the controls

WHAT CHANGES FOR THE CUSTOMER

- Specific weekly offers with a clear store or delivery option
- Prices and dates checked against the catalogue before every send
- At least one food staple in every message; Kazakh text reviewed by a native speaker

EVIDENCE (WEEK 6)

- Пайда page: “up to 60%” promised, 53% maximum, no next step [23]
- AI draft met 4 of 4 criteria vs 1 of 4 for the published text
- Checking still caught an unconfirmed Magnum Go offer and a non-food deal

GOVERNED BY

- Public promo list only; segments stay in the CRM [7]
- Automatic price check + brand-manager sign-off
- AI-use notice per legal’s decision under the AI Law [27]

DELAYED

Purchase-history personalisation: Carrefour was fined €2.25 m partly for keeping loyalty data of 28 m inactive customers [26].

Sources: Weekly Records 6–7, A3; [7], [23], [26], [27]. Budget-conscious households at 9.8% inflation [4]; 2 m+ Magnum Club members, company claim [24].

The model moves people into review roles: retrain now, redeploy later, hire no AI specialists

Action	Who	When	Capability built
Retrain	15 pilot staff: category manager, planner, store leads, copywriters, Kazakh reviewer, CRM and brand managers	Months 0–3	Forecast-error and margin reading; the four data checks; fallback drill
Retrain	Finance analysts	Months 4–8	Four-check discrepancy audit (Week 8)
Retrain	Planners in each scaled branch	Months 9–12	Reviewing and overriding forecasts
Hire or redeploy	One data steward: 0.5 FTE, then 1 FTE (from IT)	Months 0–12	Data reconciliation and quality rules
Redeploy	Time saved on manual reconciliation and drafting	Months 9–12	Shelf checks and customer service, as at Carrefour [37]
Recruit into review	Magnum Start graduates [36]	Ongoing	Planner and analyst roles in the review layer

Why: drafting is the most exposed entry task [40]; seven in ten employers treat analytical thinking as essential [41]. No roles are cut in year 1.

Sources: Weekly Record 9 capability map, A4; [36], [37], [40], [41].

Five controls apply in every function, so a failure stops one output, not the business

DATA BOUNDARY
No personal data in external AI; monthly prompt audit [7]

NAMED SIGN-OFF
Recorded approver for every order, message and figure [16]

PRE-RELEASE CHECKS
Block stale inputs, price mismatches, unflagged discrepancies

BIAS RULES
Staple rule; results by language; WAPE by branch

FALLBACK
Manual route in every function, drilled quarterly

Risk	Replenishment	Promotions	Finance reporting	Catalogue search
Privacy	Aggregate sales only	Public promo list only; CRM segments	Internal ledgers, no customer data	No account data in queries
Security	Licensed seats; export logs	Licensed seats	Restricted ledger extracts	Vendor review first
Bias	WAPE by branch	Staple rule; results by language	Same checks every month	Error gap by language
Reliability	Input checks; manager approval	Price check; sign-off	Four checks; sign-off	Checked-availability filter
Accountability	Category manager	Brand manager	Finance manager	Service manager
Escalation	Supply-chain lead; IT	Legal (labelling); DPO	Head of finance	Digital product lead
Fallback	Physical-count order	Message template	Manual report	Ordinary search, staff

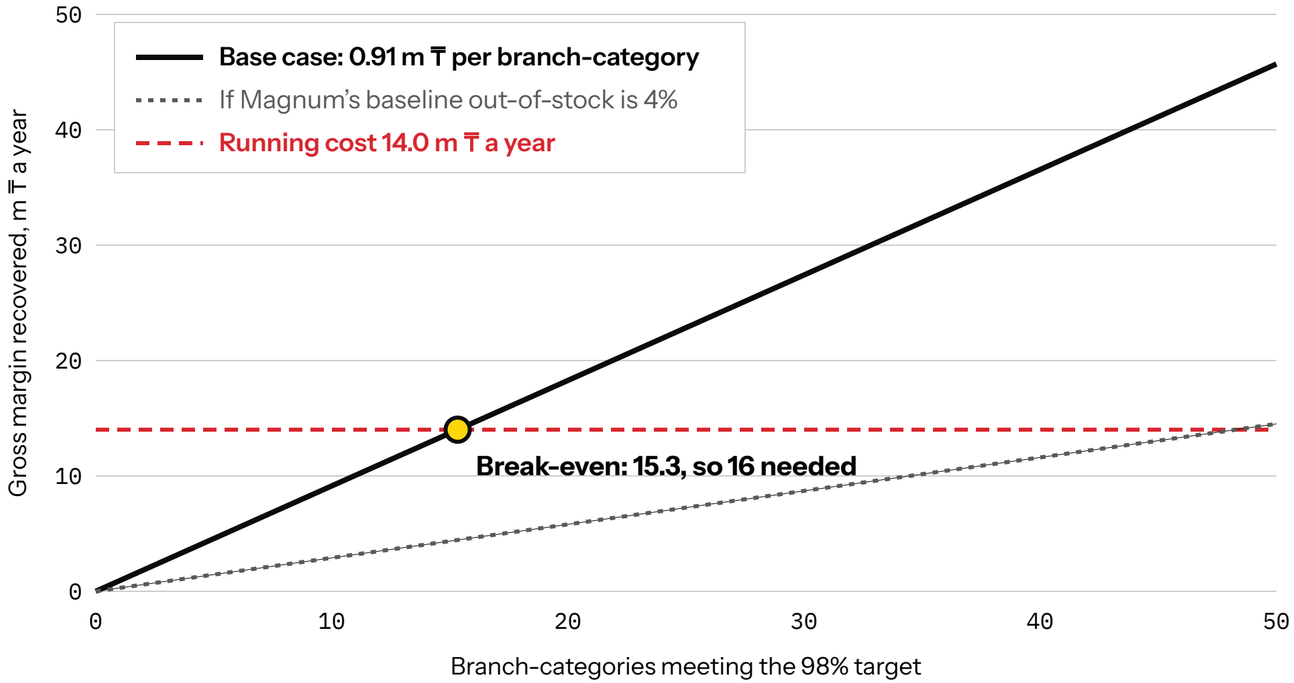
AI-assisted visual Sources: Week 7 risk register extended to the whole model; [7], [16], [27]. Residual risks: human error under time pressure; AI-labelling question [27].

Year 1 buys evidence; the case pays back from about 16 branch-categories at target

Cost line	m ₸
AI assistant licences	1.09
Data steward (redeployed or hired)	6.81
IT work: sales and stock exports	1.28
IT work: catalogue search test	0.64
Training and fallback drills	0.86
Review time for forecasts	1.47
Legal and personal-data review	0.29
Contingency 20%	2.49
Total, 12 months	14.91

3.4
m ₸ year-1 benefit if every gate passes

124
m ₸ a year net at one category × 230 branches



Benefit per branch-category = 1.97 bn ₸ branch sales × 8% category share × (8.3% − 2%) × 4%/8.3% lost sales × 19.1% margin. Wages Q2 2026 [44]; $20 a seat [45] at 452.59 ₸/$ [46].

AI-assisted visual Sources: own calculation; [2], [31], [36], [43]–[46]. Category share (8%) and Magnum's baseline out-of-stock rate are assumptions measured in months 0–3.

Nine KPIs, each with a baseline, a target and a named owner

KPI	Baseline	Target	Owner	Type
Shelf availability, pilot category	Measured in months 0–3 (placeholder 91.7%)	≥98% by month 8	Category manager	Lagging
Forecast error (WAPE)	Months 0–3	−10% relative	Planner	Leading
Waste value, pilot category	Months 0–3	−10%	Store manager	Lagging
Price-check pass before send	Week 6 sample: one untrue claim	100%	Brand manager	Leading
Price complaints a week	4 weeks before the pilot	No increase	CRM manager	Lagging
Orders from message links	Current messages	At least baseline	CRM manager	Lagging
Personal data in prompt audits	No audit today	0	Person responsible for personal data	Leading
Unflagged discrepancies in KPI report	Not measured	0	Finance manager	Leading
Pilot staff trained and drilled	0 of 15	15 of 15 by month 3	HR lead	Leading

Baselines marked “months 0–3” are measured before any AI output is used. Leading indicators warn early; lagging indicators confirm results.

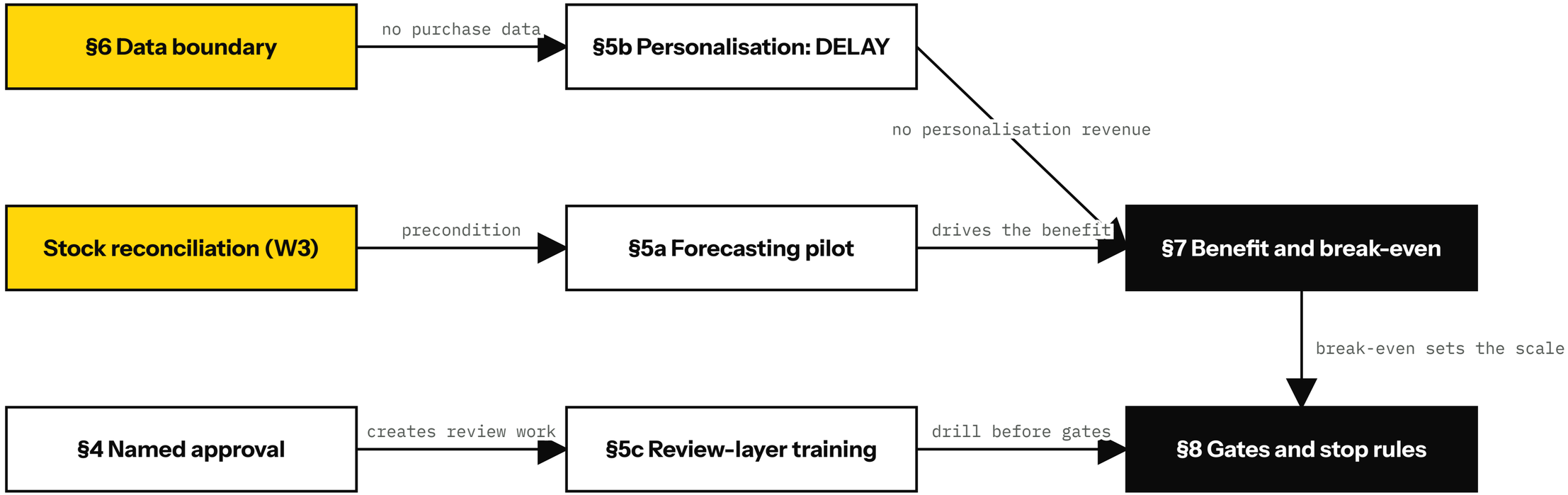
Sources: Weekly Records 5–8; targets are proposals and decision gates, not Magnum results.

Nine gated initiatives over three horizons, each with all seven elements

#	H	Action	Owner	KPI: baseline → target	Dependency	Cost and basis	Risk → control	Decision gate
1	0–3	Set the governance controls and data boundary: approval log, prompt-log audit, retention rule, AI-use notice draft, legal decision on labelling	Person responsible for personal data	Personal data found in monthly prompt audits: no audit today → 0 in every audit; approval log 100% complete	None; starts now	100 h legal and DPO time and 2 seats ≈ 0.51 m ₸ ([44]–[46])	Controls exist only on paper → audit results reported monthly to the head of marketing	Continue if audits are clean; stop all AI drafting if personal data appear in prompts
2	0–3	Reconcile stock and promotion records in one store and category; measure baselines; train 15 pilot staff	Category manager with the data steward	Recorded vs shelf stock match: unknown → ≥98% of SKUs; baselines for availability, WAPE and waste; training 0 → 15	Data steward assigned; IT sales and stock export	Data steward 0.5 FTE and IT 0.25 FTE for 3 months, 240 training hours ≈ 2.56 m ₸ ([44])	Records do not reconcile → forecasting moves to DELAY (Week 3 rule)	Start forecasting if the match is ≥98% and the fallback drill is passed; otherwise extend reconciliation
3	0–3	Pilot AI-drafted Пайда messages for four weeks behind an automatic price check and named sign-off	Brand manager	Price-check pass before send: Week 6 sample had one untrue claim → 100%; price complaints: 4-week baseline → no increase	Data boundary live (initiative 1)	4 licence seats ≈ 0.43 m ₸ a year ([45], [46]); review replaces drafting time	Wrong price or claim → automatic check, sign-off, template fallback	Extend if 100% pass and complaints flat; stop if a wrong price reaches customers
4	4–8	Pilot AI demand forecasting with draft quantities in one store and category; the manager approves every order	Category manager	Shelf availability: month-3 baseline → ≥98%; WAPE −10%; waste value −10%	Initiative 2 gate passed	Review 0.25 FTE, data steward 0.5 FTE, IT 0.25 FTE and 2 seats ≈ 3.55 m ₸ ([44]–[46])	Confident but wrong quantities → input checks, approval, DELAY trigger	Scale if all three targets are met; stop if availability is below baseline two weeks running
5	4–8	Extend AI messaging to every week with Kazakh review, the food-staple rule and the AI notice	CRM manager	Orders from message links: pre-pilot baseline → at least baseline; weeks with a staple: 100%	Initiative 3 gate; legal labelling decision	Within existing seats; Kazakh review about 2 h a week in the current copy role	Drift to non-food deals or weaker Kazakh copy → staple rule, results by language	Continue after 8 weeks at or above baseline; stop if price complaints rise
6	4–8	Introduce an AI-assisted monthly KPI report with the four-check discrepancy audit	Finance manager	Unflagged discrepancies per report: not measured → 0; preparation time: month-4 baseline → −20%	Data steward controls ledger extracts	2 seats ≈ 0.22 m ₸; analyst time neutral ([45], [46])	Period, rounding, inflation or scope errors (Week 8) → four checks and sign-off	Continue after three clean months; stop if an unflagged error reaches management
7	9–12	Scale forecasting to ten branches in the pilot category; plan a second category for year 2	Supply-chain lead	Availability ≥98% in at least 7 of 10 branches; WAPE −10%	Precondition: pilot targets met and each branch reconciled; data steward at 1 FTE	Review 0.5 FTE, data steward 1 FTE, 80 training hours ≈ 4.52 m ₸ ([44])	Data quality differs by branch → reconcile each branch before go-live	Add a second category if break-even (≈16 branch-categories) is in reach; stop if fewer than 7 branches hit target
8	9–12	Test Kazakh and Russian catalogue search on items with checked availability	Digital product lead	Successful searches vs ordinary search: measured in test → higher in both languages	Precondition: reliable availability data in pilot branches; privacy review	0.25 FTE ICT × 3 months ≈ 0.64 m ₸ ([44])	Unavailable or unsuitable items suggested → availability filter, staff help kept	Expand if success beats ordinary search in both languages; stop on any harmful recommendation
9	9–12	Review personalisation readiness without launching it	Person responsible for personal data with the CRM manager	Retention rule approved, inactive-member data purged, opt-out live: 0 of 3 → 3 of 3	Precondition: labelling decision under the AI Law; six months of clean audits	Within the 100 h legal and DPO time of initiative 1	Scope creep into purchase histories → board approval required	Propose a year-2 pilot only if all three controls are live and audited clean

Sources: Weekly Records 3–9, A1–A4; costs from §7 (subtotal 12.43 m ₸ + 20% contingency); wages [44], licences [45], [46].

Every gate rests on two foundations: the data boundary and reconciled stock records



AI-assisted visual Source: author's integration of §4–§8. Yellow = foundations started in months 0–3; black = where the sections meet in money and decisions.

AI was right on arithmetic and wrong on judgment; every figure was checked against a second source

Claim or output	Independent check	Outcome
AI interpretation: “2025: 453.13 bn ₸”	Forbes 2025 ranking uses 2024 revenue [31]	Corrected: fiscal years shifted back by one
AI: revenue per employee rising = “higher productivity”	2023 headcount is a round 10,000, as on Magnum’s careers site [32], [36]	Rejected for 2023
Пайда page: “up to 60%”	Saved 51-product list: maximum 53% [23]	Claim untrue that week; used as baseline
AI draft: “order via Magnum Go”	Пайда price validity on Magnum Go unconfirmed [25]	Flagged; human gate added
Counterargument figure: 13% decline for young workers	Revised Stanford paper: 19% below trend [39]	Updated to the revised figure
Business-case wage, licence and FX inputs	BNS Q2 2026 [44]; OpenAI pricing [45]; NBK rate [46]	Verified on 25 Oct 2026

Standard used throughout: isolate one number, find an independent source, compare period, units and entity, record the outcome even when it disagrees. AI use at Level 3 is itemised in the AI-use declaration.

Sources: Evidence Folder [23], [25], [31], [32], [36], [39], [44]–[46]; Week 8 discrepancy audit; A4.

Three assumptions could break the plan; the governance design survives all seven shocks

CONTINGENT 1

Baseline out-of-stock is high enough. Below ≈3%: stop scaling; keep messaging and finance.

CONTINGENT 2

Stock records reconcile. If not: forecasting moves to DELAY (Week 3 rule).

CONTINGENT 3

One AI tool at ≈$20 a seat. A cost shock raises break-even; templates keep campaigns running.

Shock category	Still holds	Changes	Would abandon	Need to know first
Regulatory change	Data boundary, sign-off	Hosting of AI tools; notice wording	Any external tool touching aggregated data, if banned	Exact scope of the new rule
Cost shock	Governance, pilots	Break-even rises; scaling slows	Messaging AI if cost exceeds time saved	New price and contract terms
Data-quality failure	Messaging, finance checks	Forecasting moves to DELAY	Scaling in year 1	Which records fail, and why
Partner or supplier loss	Templates and manual routes	Switch to another licensed tool	Vendor-specific features	Data export and switching time
Demand change	Controls and KPIs	Retrain forecasts; message channels	Year-1 scale targets	Size and duration of the shift
Capability loss	Templates, manual orders	Pause reconciliation; delay pilot	Parallel initiatives	Who can cover the data steward
System incident	Stop rule and fallback	Pause the AI function; root cause	The failing use until fixed	How the error passed the checks

Source: author’s analysis of the published shock categories against §4–§8. Robust: named sign-off, data boundary, manual fallbacks.

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Numbers follow the cumulative Evidence Folder (complete list in KNIAZEV_AIB-2602_BA2026_Capstone_Evidence.pdf).